Solutions to Chapter 1 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

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Exercise 1.1

Let $l(x) = ax + b$. For $a \neq 1$ the only fixed point is $p = b/(1-a)$, and $l' = a$. It is attracting if $|a| < 1$ and repelling if $|a| > 1$, for any b. If $a = 1$: every point is fixed when $b = 0$, and there are no fixed points when $b \neq 0$. If $a = -1$: $p = b/2$ is neutral and every other point has period 2.

Exercise 1.2

(a) $f(0) = 0$ and $f'(0) = 1$. For small $x > 0$, $f(x) = x - x^2 < x$, so the orbit decreases, stays positive, and converges to 0. For $x < 0$, $f(x) < x$, so the orbit moves away. Thus 0 attracts from the right and repels on the left.

(b) $\tan 0 = 0$ and $g'(0) = 1$. Since $\tan x > x$ for $0 < x < \pi/2$ and tan is odd, points on both sides move away from 0: 0 is repelling.

(c) $h(x) = x - x^3$: $h'(0) = 1$ and $|h(x)| < |x|$ near 0, so 0 is attracting.

(d) $h(x) = x + x^3$: $h'(0) = 1$ and $|h(x)| > |x|$ near 0, so 0 is repelling.

Exercise 1.3

$f(x) - x = x^3 = 0$ only at $x = 0$, so 0 is the only fixed point. For $x > 0$, $f(x) > x$, so the orbit increases; with no fixed point to converge to, it tends to $+\infty$. For $x < 0$ it tends to $-\infty$. Hence 0 is a source.

Exercise 1.4

Write $x = \sin^2 \varphi$. Then $G(x) = \sin^2(2\varphi)$, so $G^3(x) = x$ means $\sin^2(8\varphi) = \sin^2 \varphi$, i.e. $\varphi = k\pi/7$ or $\varphi = k\pi/9$. For $\varphi \in [0, \pi/2]$ this gives eight distinct points (all the roots of the degree-8 polynomial $G^3(x) - x$), in increasing order:

$$0,\ 0.1170,\ 0.1883,\ 0.4132,\ 0.6113,\ 0.75,\ 0.9505,\ 0.9698.$$

(a) $x_6 = 3/4$ (that is, $\varphi = \pi/3$).

(b) $\{x_2, x_4, x_8\} = \{\sin^2\frac{\pi}{9}, \sin^2\frac{2\pi}{9}, \sin^2\frac{4\pi}{9}\}$ is one orbit, since doubling the angle again gives $\sin^2\frac{8\pi}{9} = \sin^2\frac{\pi}{9}$. The other orbit is $\{x_3, x_5, x_7\} = \{\sin^2\frac{\pi}{7}, \sin^2\frac{2\pi}{7}, \sin^2\frac{3\pi}{7}\}$, since $\sin^2\frac{4\pi}{7} = \sin^2\frac{3\pi}{7}$ and $\sin^2\frac{8\pi}{7} = \sin^2\frac{\pi}{7}$.

Exercise 1.5

Solving $f(f(x)) = x$ gives $8x(x-3)(x^2-2x-1) = 0$. The roots $0, 3$ are fixed points, so the period-two orbit is $\{1+\sqrt{2},\, 1-\sqrt{2}\}$. For these points p, q we have $p+q = 2$ and $pq = -1$, so

$$(f^2)' = f'(p)f'(q) = (4p-5)(4q-5) = 16pq - 20(p+q) + 25 = -31.$$

Since $|-31| > 1$, the orbit is a **source**.

Exercise 1.6

$f(x) = 2x \pmod 1$, $L = [0, \frac{1}{2}]$, $R = [\frac{1}{2}, 1]$.

(a) Each of L, R is stretched onto $[0,1]$. The itinerary chart has $LL = [0, \frac{1}{4}]$, $LR = [\frac{1}{4}, \frac{1}{2}]$, $RL = [\frac{1}{2}, \frac{3}{4}]$, $RR = [\frac{3}{4}, 1]$; in general each itinerary of length n is an interval of length 2^{-n} (the itinerary is the binary expansion of x).

(b) The transition graph has arrows $L \to L$, $L \to R$, $R \to L$, $R \to R$.

(c) Fix x and $\varepsilon > 0$. Choose n with $2^{-(n+1)} < \varepsilon$ and let $y = x \pm 2^{-(n+1)}$ (sign chosen so $y \in [0,1]$). Then $f^n(y) \equiv f^n(x) + \frac{1}{2} \pmod 1$, so $|f^n(y) - f^n(x)| = \frac{1}{2}$. Every point has arbitrarily close neighbors that end up $\frac{1}{2}$ apart.

Exercise 1.7

Tent map T, $L = [0, \frac{1}{2}]$, $R = [\frac{1}{2}, 1]$.

(a) Each of L, R is mapped onto $[0,1]$, with R reversed. The itinerary chart has $LL = [0, \frac{1}{4}]$, $LR = [\frac{1}{4}, \frac{1}{2}]$, $RR = [\frac{1}{2}, \frac{3}{4}]$, $RL = [\frac{3}{4}, 1]$. The transition graph is the same as in 1.6. For sensitive dependence, let $\varepsilon > 0$ and choose n with $2^{-n} < \varepsilon$. T^n is linear with slope $\pm 2^n$ on each of 2^n intervals of length 2^{-n}, and maps each onto $[0,1]$. Let J be the interval containing x. Its endpoints map to 0 and 1, so at least one endpoint y has $|T^n(x) - T^n(y)| \geq \frac{1}{2}$, while $|x - y| \leq 2^{-n} < \varepsilon$.

(b) T^n has 2^n fixed points, since each of its 2^n linear pieces crosses the diagonal once. Removing points of lower period (Möbius inversion) gives:

Period	1	2	3	4	5	6	7	8	9	10
Points	2	2	6	12	30	54	126	240	504	990
Orbits	2	1	2	3	6	9	18	30	56	99

These counts are identical to those for G. The differences lie in the locations of the points (rationals for T; for G their images under $C(x) = (1 - \cos \pi x)/2$, e.g. values $\sin^2(r\pi)$) and at the fixed point 0, where $T'(0) = 2$ but $G'(0) = 4$. Every other periodic orbit of period k has multiplier $\pm 2^k$ for T and, by conjugacy (Theorem 3.11), also $\pm 2^k$ for G.

Exercise 1.8

For each n, $f^n(x) = x$ is a polynomial equation of degree 2^n, so it has finitely many roots. For each m, n, $f^{m+n}(x) = f^m(x)$ is a nonzero polynomial equation (degree $2^{m+n} > 2^m$), so it also has finitely many roots. Hence the set of fixed, periodic and eventually periodic points is a countable union of finite sets, so it is countable. Since $[0, 1]$ is uncountable, some points are none of these.

Exercise 1.9

Let $g(x) = (x + 2)/(x + 1)$. Fixed points satisfy $x^2 = 2$, so $x = \pm\sqrt{2}$. Direct algebra gives

$$g(x) - \sqrt{2} = \frac{(1 - \sqrt{2})(x - \sqrt{2})}{x + 1}, \qquad g(x) + \sqrt{2} = \frac{(1 + \sqrt{2})(x + \sqrt{2})}{x + 1}.$$

Dividing,

$$\frac{x_{n+1} - \sqrt{2}}{x_{n+1} + \sqrt{2}} = \lambda \frac{x_n - \sqrt{2}}{x_n + \sqrt{2}}, \qquad \lambda = \frac{1 - \sqrt{2}}{1 + \sqrt{2}} = -(3 - 2\sqrt{2}), \quad |\lambda| \approx 0.17 < 1.$$

(a) For $x_0 \geq 0$, the first identity gives $|x_{n+1} - \sqrt{2}| \leq (\sqrt{2} - 1)|x_n - \sqrt{2}|$, so $L = \sqrt{2}$.

(b) For any x_0 whose orbit is defined, $r_n = \frac{x_n - \sqrt{2}}{x_n + \sqrt{2}} = \lambda^n r_0 \to 0$, so $x_n \to \sqrt{2}$ unless $x_0 = -\sqrt{2}$. The exceptional negative x_0 are:

- $x_0 = -\sqrt{2}$, which is fixed, so the limit is $-\sqrt{2} \neq L$;
- the points that reach -1 after finitely many steps, where the next term is undefined and the limit does not exist. These are the preimages $g^{-n}(-1)$, $n \geq 0$, with $g^{-1}(y) = \frac{2-y}{y-1}$: namely -1, $-\frac{3}{2}$, $-\frac{7}{5}$, $-\frac{17}{12}$, $-\frac{41}{29}, \ldots$, which accumulate at $-\sqrt{2}$.

Exercise 1.10

$g(x) = 3.05\,x(1 - x)$, $a = 3.05$. Fixed points: 0 and $p = 1 - 1/a = 41/61 \approx 0.672$. $g'(0) = 3.05 > 1$, so 0 is a source. $g'(p) = 2 - a = -1.05$, so p is also a source. Period-two points: $x = \dfrac{(a + 1) \pm \sqrt{(a + 1)(a - 3)}}{2a}$, i.e. $x = 36/61 \approx 0.590$ and $45/61 \approx 0.738$. The multiplier is $(g^2)' = a^2(1 - 2p)(1 - 2q) = -a^2 + 2a + 4 = 0.7975$. Since $|0.7975| < 1$, the period-two orbit is a **sink**.

Exercise 1.11

(a) Suppose f is not monotone. Then there are $x_1 < x_2 < x_3$ with $f(x_2)$ larger (or smaller) than both $f(x_1)$ and $f(x_3)$. Pick c strictly between $f(x_2)$ and the larger (or smaller) of $f(x_1), f(x_3)$. By the intermediate value theorem c is attained once in (x_1, x_2) and once in (x_2, x_3), contradicting one-to-one. So f is strictly monotone: increasing for all x or decreasing for all x.

(b) $h = f^2$ is increasing (a composition of two increasing maps, or of two decreasing maps, is increasing). If $x_0 \le x_1 = h(x_0)$, applying h gives $x_1 \le h(x_1) = x_2$, then $x_2 \le x_3$, and so on, so the orbit is nondecreasing. If $x_0 \ge x_1$, the same argument gives a nonincreasing orbit.

(c) A monotone sequence is either unbounded, in which case it diverges to $+\infty$ (nondecreasing) or $-\infty$ (nonincreasing), or bounded, in which case it converges to some L. By continuity $h(L) = L$, so L is a fixed point of f^2.

(d) Both the even terms x_{2k} and the odd terms $x_{2k+1} = f(x_{2k})$ are orbits of f^2. If $x_{2k} \to p$ with $f^2(p) = p$, then $x_{2k+1} \to f(p)$ and $f(f(p)) = p$. So every orbit of f either diverges to $\pm\infty$, or converges to a fixed point of f, or converges to a period-two orbit of f. There is no chaos and no other periods.

Exercise 1.12

$h(x) = axe^{-x}$.

(a) A nonzero fixed point satisfies $ae^{-x_0} = 1$, so $x_0 = \ln a$. Then $h'(x) = ae^{-x}(1 - x)$ gives $h'(x_0) = 1 - x_0$. This is 0 iff $x_0 = 1$, so $a = e$ and $x_0 = 1$.

(b) For $a = e$ and $x_0 = 0.1$ the orbit is approximately $0.1,\ 0.246,\ 0.523,\ 0.843,\ 0.986,\ 0.99995, \cdots \to 1$, converging very rapidly (superstable). For $a = 1.5e \approx 4.077$ the fixed point is $\ln a \approx 1.405$ with $h' = 1 - \ln a \approx -0.405$, so it is still a sink, but convergence is slower and oscillating: $0.1,\ 0.369,\ 1.040,\ 1.499,\ 1.365,\ 1.421,\ 1.399, \cdots \to 1.405$.

(c) For $a > 1$ the positive fixed point is $x^* = \ln a$, with $h'(x^*) = 1 - \ln a$. It is a sink iff $|1 - \ln a| < 1$, i.e. $0 < \ln a < 2$. So the range is $1 < a < e^2 \approx 7.39$. (At $a = 1$ the only fixed point is 0, and at $a = e^2$ we get $h' = -1$.)

Exercise 1.13

Let f be smooth on $[0, \infty)$ with $f \ge 0$, $f(0) = 0$, $f(p) = p$ for some $p > 0$, and f' decreasing (so f is strictly concave).

Step 1: $f' \ge 0$ everywhere. If $f'(x_1) < 0$ for some x_1, then $f' < f'(x_1) < 0$ for all $x > x_1$, so $f \to -\infty$, contradicting $f \ge 0$. Since f' is strictly decreasing and ≥ 0, in fact $f' > 0$, so f is increasing.

Step 2: sign of $f(x) - x$. Let $g(x) = f(x) - x$, which is strictly concave with $g(0) = g(p) = 0$. For $x \in (0, p)$, strict concavity gives $g(x) > 0$. For $x > p$, write $p = tx$ with $t \in (0, 1)$; then $0 = g(p) > t\, g(x)$, so $g(x) < 0$. Thus $f(x) > x$ on $(0, p)$ and $f(x) < x$ on (p, ∞).

Step 3. If $0 < x_0 < p$, then since f is increasing, $x_0 < x_1 = f(x_0) < f(p) = p$, and inductively x_n increases and is bounded by p. If $x_0 > p$, then $p = f(p) < x_1 = f(x_0) < x_0$, and x_n decreases and is bounded below by p. In both cases the orbit converges to a fixed point in $(0, \infty)$; the only one is p. Hence every $x_0 > 0$ converges to p.

Exercise 1.14

$f(x) = x^2 + x$. Fixed points: $x^2 = 0$, so only $x = 0$. Since $f(x) - x = x^2 \geq 0$, orbits never decrease.

- $x > 0$: the orbit increases and, with no fixed point above, tends to $+\infty$.
- $-1 < x < 0$: $f(x) = x(x+1) < 0$ and $f(x) > x$, so the orbit increases while staying negative, and converges to the fixed point 0.
- $x = -1$: $f(-1) = 0$, so the orbit is fixed at 0 from then on.
- $x < -1$: $f(x) = x(x+1) > 0$, so the next point is positive and the orbit tends to $+\infty$.

So points in $[-1, 0]$ go to 0, and all other points go to $+\infty$.

Exercise 1.15

Let $\theta = \arccos(1 - 2x_0)$, so $\cos\theta = 1 - 2x_0$ and $x_0 = \frac{1}{2} - \frac{1}{2}\cos\theta$; the formula holds for $n = 0$. Suppose $x_n = \frac{1}{2} - \frac{1}{2}\cos(2^n\theta)$, and write $c = \cos(2^n\theta)$. Then

$$x_{n+1} = 4x_n(1 - x_n) = 4 \cdot \frac{1-c}{2} \cdot \frac{1+c}{2} = 1 - c^2 = \sin^2(2^n\theta) = \frac{1}{2} - \frac{1}{2}\cos(2^{n+1}\theta),$$

using $\cos 2\phi = 1 - 2\sin^2\phi$. By induction, $x_n = \frac{1}{2} - \frac{1}{2}\cos\big(2^n \arccos(1 - 2x_0)\big)$.

Exercise 1.16

Condition: there is a constant $\lambda < 1$ with $|f(x) - p| \leq \lambda|x - p|$ for all $x \in (p - \varepsilon, p + \varepsilon)$. For example, it suffices that f is C^1 with $|f'(x)| \leq \lambda < 1$ on $[p - \varepsilon, p + \varepsilon]$, by the Mean Value Theorem.

Proof. Then f maps $(p - \varepsilon, p + \varepsilon)$ into itself, and by induction $|f^n(x) - p| \leq \lambda^n|x - p| \to 0$.

Exercise 1.17

Write $x = \sin^2\varphi$ with $\varphi \in (0, \pi/2)$. Then $f^i(x) = \sin^2(2^i\varphi)$. Claim: $0 < f^i(x) < \frac{1}{2}$ for $0 \leq i < k$ iff $\varphi < \pi/2^{k+1}$.

Positivity is automatic when $2^i\varphi \in (0, \pi/2)$. If $2^i\varphi \in (0, \pi/2)$, then $\sin^2(2^i\varphi) < \frac{1}{2} \iff 2^i\varphi < \pi/4$, which in turn gives $2^{i+1}\varphi < \pi/2$, so the same reasoning applies at step $i + 1$. By induction the conditions for $i = 0, \ldots, k - 1$ all hold iff the last one holds: $2^{k-1}\varphi < \pi/4$, i.e. $\varphi < \pi/2^{k+1}$.

So the set of such x is the interval $\big(0, \sin^2(\pi/2^{k+1})\big)$, whose length is

$$\sin^2\Big(\frac{\pi}{2^{k+1}}\Big) = \frac{1-\cos(\pi/2^k)}{2}.$$